

An abstract network diagram with white nodes and connecting lines on a blue background. The nodes are represented as small white spheres, and the connections are thin white lines. The network is complex and interconnected, with some nodes having multiple connections. The background is a solid light blue color.

Introduction to Network Simulator-3

Introduction to Network Simulators

- Tools used to simulate and analyze the behavior of computer networks.
- Popular Simulators:
 - NS2
 - NS3
 - OMNeT++
 - GNS3

Overview of NS2

- **Written in:** C++ with OTcl scripting interface
- **Supports:** Wired, wireless, etc.
- **Used for:** Academic research and protocol development
- **Limitations of NS2**
 - Programming Complexity
 - Scalability Issues
 - Limited Flexibility
 - Outdated Models
 - No Real Hardware Integration
 - Graphical Visualization
 - Performance Limitations

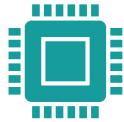
Why opt for NS3?



Unified
Programming (C++
& Python)



Modern
Architecture



Scalability &
Performance



Real-World
Integration



Advanced
Visualization



Community and
Documentation



Model Support



Open-source and
Actively Developed

Introduction to NS3

- **NS3 (Network Simulator 3)** is an open-source, discrete-event network simulator.
- **Released:** 2008 (as a replacement for NS2)
- **Languages:** C++ (core), Python (optional scripting)
- Designed **for research and educational purposes**, especially in wireless and wired network simulation.
- **Open-source** and actively maintained
- Developed from scratch to **replace NS2** with modern features and better architecture.

Features of NS3

- **NS3 allows simulation of:**
 - Realistic Internet stacks (IPv4/IPv6)
 - Wireless protocols (Wi-Fi, LTE, 5G, ZigBee)
 - Mobile networks (MANETs, VANETs)
 - IoT and emerging technologies
- **Tools:**
 - NetAnim, PyViz for visualization
 - Tracing and logging utilities

Installation Prerequisites

- Ubuntu/Linux environment or a VM installed
- C++ & Python are installed in the system.

Commands for Installation

- `sudo apt update && sudo apt install -y \`
- `g++ python3 python3-dev python3-setuptools pkg-config sqlite3 \git python3-gi python3-gi-cairo python3-pygraphviz gir1.2-goocanvas-2.0 \gir1.2-gtk-3.0 ipython3 \openmpi-bin openmpi-common openmpi-doc libopenmpi-dev \autoconf cvs bzip2 unrar \tcpdump wireshark libxml2 libxml2-dev`
- `sudo apt install qtbase5-dev qtchooser qt5-qmake qtbase5-dev-tools`

Installation of NS-3

- Download the ns-allinone-3.42
- Copy the file in the home directory, extract.
- Go to ns-allinone folder: `cd ns-allinone-3.42`
- `./build.py --enable-examples --enable-tests`
- Go to ns-3.42 folder
- `cd build`
- `cmake ..`
- `make`
- To check whether the simulator is successfully installed or not :
 - Go to the scratch folder: `./ns3 run scratch/scratch-simulator`

Installation of NS-3

- For the Visualization:
 - `sudo apt-get update`
 - Go to the ns-allinone-3.42
 - `cd netanim-3.109/`
 - `make clean`
 - `Make`
 - `qmake NetAnim.pro`
 - To run NetAnim: `./NetAnim`

- Install dependencies
 - `sudo apt update`
 - `sudo apt install qt5-default qtscript5-dev libqt5svg5-dev build-essential`
- Build NetAnim manually
 - `cd ~/ns-allinone-3.42/netanim-3.109`
 - `qmake NetAnim.pro`
 - `make`
- Run NetAnim
 - `./NetAnim`

Understanding the ns-3 Directory Structure



- ns-3.xx/: core source code and modules
- Build/ : Build output directory
- Netanim-3.xxx/ : Network Animator module
- examples/ (inside ns-3.xx/): sample simulation program
- Scratch/: Space for user scripts and simulations
- src/: Drivers, protocols, network models
- doc/ : User and Developer documentation